Started on	Tuesday, 29 September 2026, 6:05 PM
State	Finished
Completed on	Tuesday, 29 September 2026, 6:25 PM
Time taken	20 mins 6 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Problem Statement:

Given a sorted array of integers say arr[] and a number x. Write a recursive program using divide and conquer strategy to check if there exist two elements in the array whose sum = x. If there exist such two elements then return the numbers, otherwise print as “No”.

Note: Write a Divide and Conquer Solution

Input Format

First Line Contains Integer n – Size of array

Next n lines Contains n numbers – Elements of an array

Last Line Contains Integer x – Sum Value

Output Format

First Line Contains Integer – Element1

Second Line Contains Integer – Element2 (Element 1 and Elements 2 together sums to value “x”)

Answer: (penalty regime: 0 %)

```c
#include<stdio.h>
int binarysearch(int arr[],int low,int high,int key)
{
    if(low > high)
     return-1;

    int mid=(low+high)/2;

    if(arr[mid]==key)
       return mid;
    else if(arr[mid]>key)
       return binarysearch(arr,low,mid-1,key);
     else
       return binarysearch(arr,mid+1,high,key);
}

int findpair(int arr[],int n,int x,int i)
{
    if(i>=n-1)
       return 0;

    int needed=x-arr[i];
    int pos = binarysearch(arr,i+1,n-1,needed);

    if(pos != -1)
    {
        printf("%d\n",arr[i]);
        printf("%d\n",arr[pos]);
        return 1;
    }

    return findpair(arr,n,x,i+1);
}
int main()
{
    int n,x;
    scanf("%d",&n);

    int arr[n];

    for(int i=0;i<n;i++)
       scanf("%d",&arr[i]);

    scanf("%d",&x);

    if(!findpair(arr,n,x,0))
       printf("No");

    return 0;


}
```

```
52 |}
```

	Input	Expected	Got	
✔	4 2 4 8 10 14	4 10	4 10	✔
✔	5 2 4 6 8 10 100	No	No	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.